

Sofia Nikiforova

Applied AI/ML Engineer & Data Scientist

NLP · LLMs · 8+ Years Python

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Applied AI researcher and ML engineer specializing in NLP, LLMs, and Generative AI, with 8+ years of professional Python experience and a Ph.D. in Computational Linguistics/NLP.

I design, build, evaluate, and improve AI systems for real-world use, spanning LLM-powered data analysis tools, information extraction, text generation, multimodal AI, and predictive modeling.

I have led projects from data acquisition and problem formulation through model design, implementation, evaluation, error analysis, and production-oriented delivery. My work places particular emphasis on understanding model behavior, building reliable evaluation workflows, and translating technical findings and stakeholder feedback into measurable system improvements.

Experience

Senior AI Researcher

Oct 2024–Present

Causify

Led applied AI project teams developing LLM-powered data analysis tools, causal inference systems, and time series prediction solutions. Owned work across data acquisition, problem formulation, exploratory analysis, model and workflow design, implementation, evaluation, stakeholder communication, and production-oriented delivery.

- Led development of an LLM-powered tool for generating, validating, and interacting with causal DAGs from datasets and metadata, with support for user feedback and on-demand causal analysis.
- Led development of a causal failure prediction framework for wind turbines, combining sensor time series data, domain knowledge, and predictive modeling to identify early indicators of equipment failure.
- Supervised and evaluated interns working on AI, data science, and causal inference projects.
- Contributed to research papers, technical reports, patent-related work, and internal tooling for code quality, testing, and reproducibility.

Postdoctoral Researcher — AI

Mar 2023–Oct 2024

Utrecht University

Developed multimodal image-text AI systems for radiology, with a focus on integrating visual and linguistic representations for clinical decision-support applications.

- Designed and implemented models combining medical imaging with structured and unstructured textual data.
- Emphasized factual accuracy, interpretability, and robustness in user-facing AI systems for a high-accuracy and safety-sensitive domain.
- Collaborated with an interdisciplinary team across AI, medicine, and data science.

Senior Data Scientist — NLP

Nov 2016–Jun 2021

Particle.One

Led the development of NLP-driven classification, knowledge extraction, and predictive modeling systems for the financial domain.

- Owned end-to-end NLP model development, from problem formulation and data acquisition to model design, evaluation, deployment, and monitoring.
- Built and evaluated systems for text classification, information extraction, and predictive analytics.
- Translated business and data problems into applied machine learning solutions.

Teaching Assistant

2021

Utrecht University

Course: Machine Learning for Human Vision and Language (MSc Artificial Intelligence).

Developed and graded assignments and supervised practical lab sessions covering state-of-the-art machine learning techniques for vision and language tasks.

NLP Developer

2017–2018

Semantic Hub

Developed NLP tools for information extraction in the life sciences/biomedical domain. Implemented pipelines for entity recognition, relation extraction, and structured data generation from heterogeneous unstructured text, including technical documents and user-generated content.

Education

Ph.D., Computational Linguistics/NLP

2018–Feb 2023

Utrecht University, Netherlands

Dissertation: *Image Captioning with External Knowledge*

Developed knowledge-grounded image caption generation models that identified relevant entities, retrieved contextual information, and incorporated that information into generated descriptions. The research focused on factual accuracy, grounding, dataset creation, and systematic evaluation across geographic and news-image domains.

M.A., Computational Linguistics

Higher School of Economics, Moscow, Russia

2016–2018

B.A., Fundamental and Applied Linguistics

Lomonosov Moscow State University (MSU), Russia

2012–2016

Selected Publications

- **Causify DataMap: Automatic Causal Probabilistic Reasoning.** *Internal Causify report, 2026.*
LLM-powered system for generating, validating, and interacting with causal DAGs from datasets and metadata, with support for user feedback and on-demand causal analysis.
- **DMCD: Semantic-Statistical Framework for Causal Discovery.** *arXiv preprint, 2026.* [arXiv:2602.20333](#)
A two-phase causal discovery framework that integrates LLM-based semantic drafting from variable metadata with statistical validation on observational data.
- **Runnable Directories: The Solution to the Monorepo vs. Multi-repo Debate.** *arXiv preprint, 2025.* [arXiv:2512.03815](#)
Software-architecture pattern that reconciles trade-offs between monorepo and multi-repo organisation of code.
- **Causify Sentinel: A Causal Failure Prediction Framework.** *Internal Causify report, 2025.*
Causal framework for predicting failures in sensor time-series systems by combining domain knowledge, predictive modeling, and causal reasoning.
- **Generating Image Captions with External Encyclopedic Knowledge.** *arXiv preprint, 2022.* [arXiv:2210.04806](#)
Image captioning approach that integrates encyclopedic knowledge into generation, with emphasis on factual accuracy.
- **Geo-Aware Image Caption Generation.** *Proceedings of COLING 2020.* [ACL Anthology](#)
Image captioning with geographic context grounding for generating location-aware photo descriptions.

Patents

- **Systems and Methods for Causal Failure Prediction in Time Series** — U.S. Patent Application No. 63/934,790

Selected Talks

- **Imperfect LLMs in High-Accuracy Fields** — NLG Summit in the Lowlands, Utrecht University (2023)
- **On factual accuracy evaluation in knowledge-aware text generation** — NLG Summit in the Lowlands, Tilburg University (2022)
- **Geo-Aware Image Caption Generation** — COLING 2020, Barcelona (Online)
- **Geographic Context for Image Captioning** — GeCKo Symposium, Barcelona (Online), 2020

Skills & Technologies

Skills

Natural Language Processing (NLP) · Large Language Models (LLMs) · Generative AI · Machine Learning · Model Evaluation · Error Analysis · Prompt Engineering · Workflow Optimization · Information Extraction · Dataset Creation · Experiment Design · Explainability · Conversational AI · Multimodal AI · Vision-Language Models · Causal Inference · Time-Series Modeling

Professional Strengths

Technical Leadership · Analytical Problem-Solving · Cross-Functional Collaboration · Stakeholder Communication · Independent Project Ownership

Technologies

Python · OpenAI API · PyTorch · scikit-learn · spaCy · pandas · NumPy · Pydantic · pytest · Git · GitHub · Jupyter · Linux · LaTeX

Languages

Russian (native) · English (fluent) · Dutch (upper-intermediate)

Location & Availability

Amersfoort, Netherlands · Dutch citizen · No sponsorship required · Available immediately